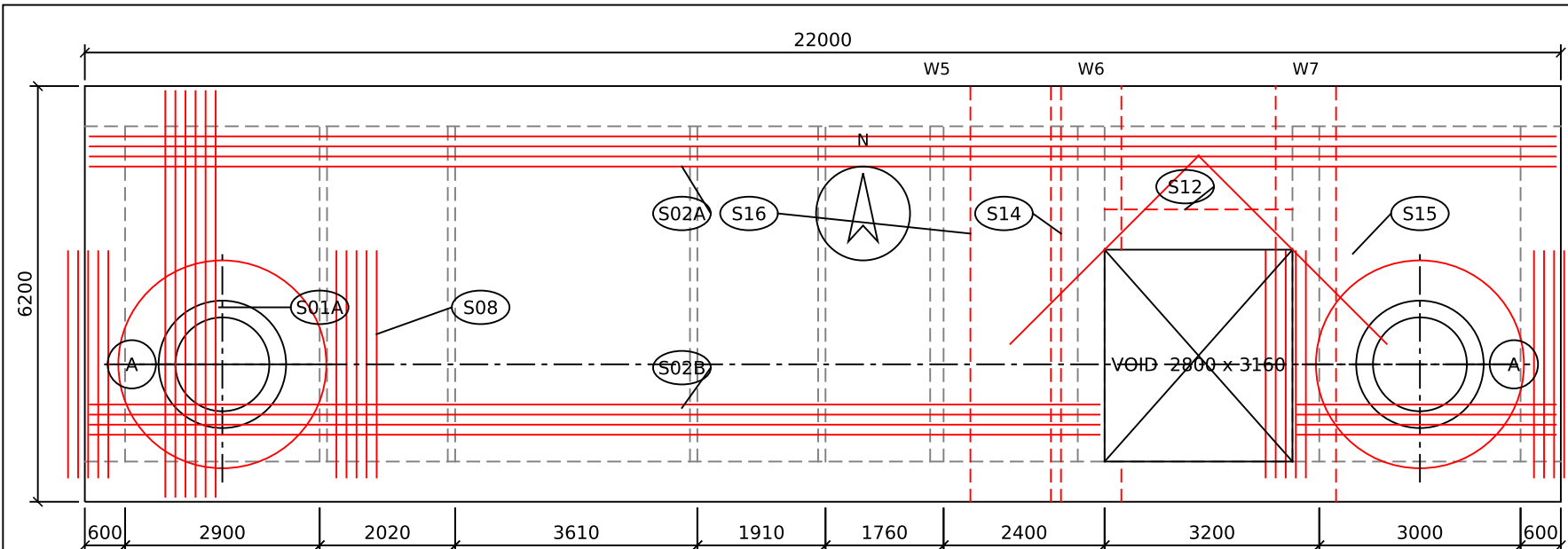
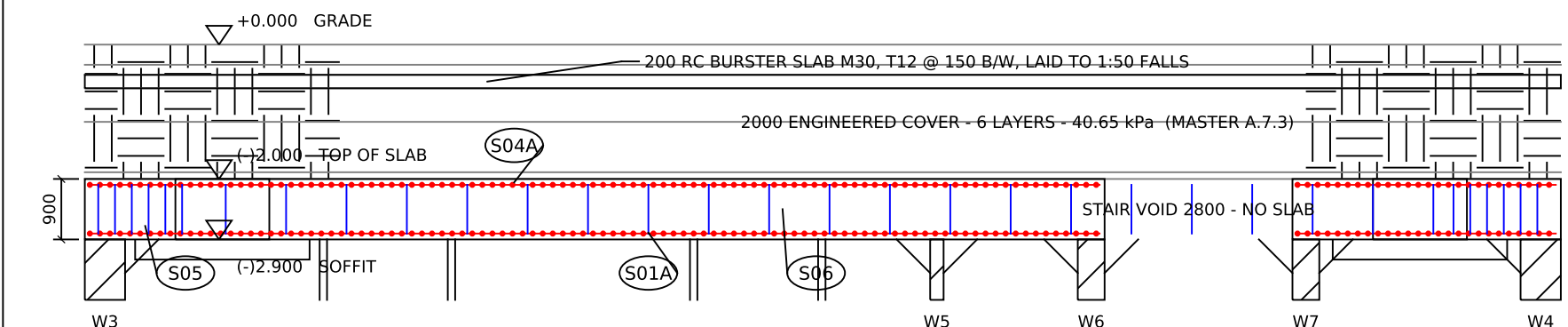
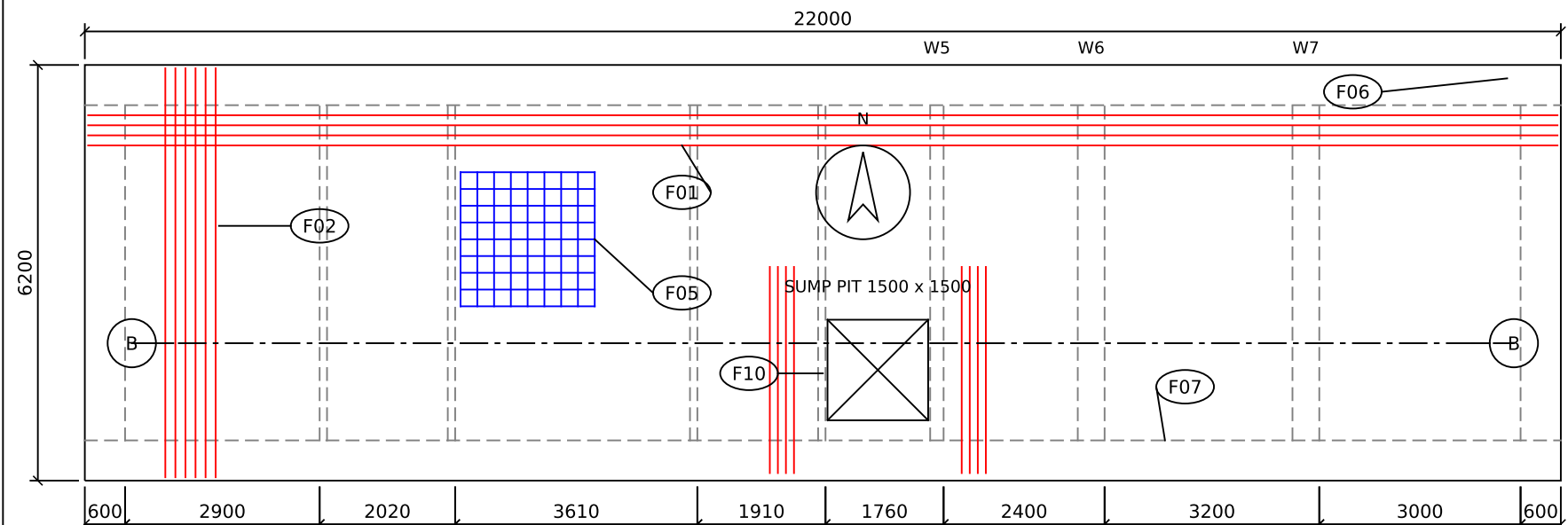




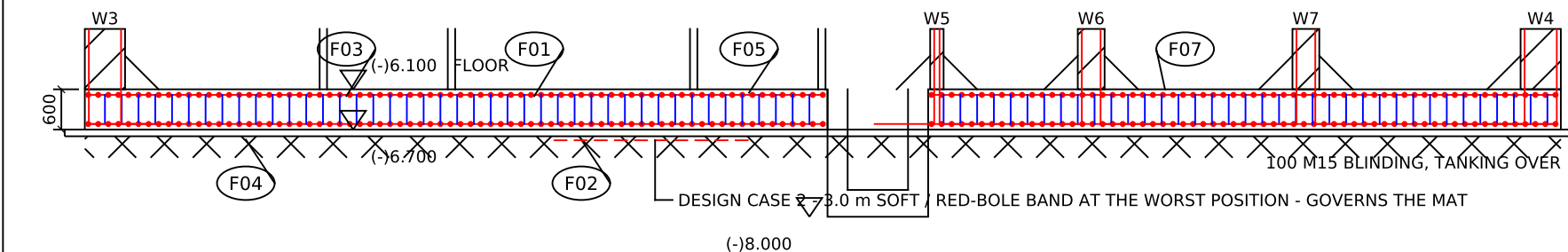
1 ROOF (PRESSURE) SLAB - REINFORCEMENT PLAN
1 : 100 SLAB 900 THK, T25 @ 150 EF EW



2 ROOF SLAB - LONGITUDINAL SECTION A-A
1 : 100 CUT AT Y 2050, LOOKING NORTH



3 MAT FOUNDATION - REINFORCEMENT PLAN
1 : 100 MAT 600 THK, T16 @ 150 EF EW



4 MAT FOUNDATION - LONGITUDINAL SECTION B-B
1 : 100 CUT AT Y 2050, LOOKING NORTH

CONSTRUCTION REQUIREMENTS THAT ARE DESIGN OUTPUTS, NOT OPTIONS

- 1 THE HAZARD IS NOT THE BASALT - IT IS THE FLOW CONTACTS. OVER-EXCAVATE ANY RED-BOLE OR VESICULAR SEAM AT FOUNDING LEVEL AND REPLACE WITH M15 LEAN CONCRETE. A single seam under the mat produces the 3.0 m differential-support case that SIZES the mat. This is a requirement, not an option.
- 2 100 mm M15 BLINDING TO (-)6.800 before any reinforcement is placed; tanking membrane on the blinding with a protection screed. 75 mm cover to the blinding on CEMENTITIOUS spacers - no plastic spacer against an earth or blinding face. Chairs at not more than 1000 c/c both ways to the top curtain.
- 3 THE T12 LINK GRID F05 IS STRUCTURAL AND IS ALSO THE CURTAIN SPACER SYSTEM - It is not optional. CONSTRUCTION JOINTS AT APPROXIMATELY 6 m WITH THE REINFORCEMENT FULLY CONTINUOUS; two waterstops and a welded Cu / galvanised EMP strap at every joint.
- 4 FLOTATION GOVERNS THE CONSTRUCTION STAGE, NOT THE FINISHED STRUCTURE. Fo5 is 0.33 with the mat alone and 1.22 with the box complete and no backfill. Continuous dewatering from the start of excavation until backfill and cover are complete; 6 No. temporary pressure-relief plugs in the mat, grouted up ONLY after backfill; backfill SYMMETRICALLY in layers. NO BAR MAY BE DISPLACED TO CLEAR A SERVICE - any clash is an RFI.

ROOF SLAB AND MAT - ELEMENT SCHEDULE

ELEMENT	THK	COVER	d	MAIN REINFORCEMENT	LINKS
ROOF SLAB - MIDSPAN	900	75	812.5	T25 @ 150 EF EW (3272 mm2/m)	T12 4L @ 250, END 1500 EACH SIDE
ROOF SLAB - MID ZONE	900	75	812.5	T25 @ 150 EF EW	T12 2L @ 300
ROOF - CANTILEVER PAD	900	75	812.5	T25 @ 150 TOP, CONTINUOUS	T12 4L @ 250 THROUGHOUT
ROOF - 1200 THICKENINGS	1200	75	-	6-T25 TOP + 6-T25 BOTTOM	T12 CLOSED @ 150
MAT FOUNDATION	600	75 / 50	517	T16 @ 150 EF EW (1340 mm2/m)	T12 ON A 250 x 250 GRID
SUMP PIT WALLS / BASE	300 / 400	50	244	T16 @ 150 EF EW	-

BAR MARK SCHEDULE - ROOF (PRESSURE) SLAB 900

MARK	BAR	SPACING	LOCATION	CUT mm	No.
S01A	T25	150	BOTTOM, TRANSVERSE (Y) - MAIN ZONES	6050	127
S01B	T25	150	BOTTOM, TRANSVERSE - CANTILEVER PAD	2365	18
S02A	T25	150	BOTTOM, LONGITUDINAL (X) - FULL LENGTH BANDS	11550	40
S02B	T25	150	BOTTOM, LONGITUDINAL - WEST OF THE VOID	8188	42
S02C	T25	150	BOTTOM, LONGITUDINAL - EAST OF THE VOID	3925	21
S03A	T25	150	TOP, TRANSVERSE (Y) - MAIN ZONES	6050	127
S03B	T25	150	TOP, TRANSVERSE - CANTILEVER PAD	2365	18
S03C	T25	150	TOP, TRANSVERSE - STRIP OVER W1 AT THE VOID	525	18
S04A	T25	150	TOP, LONGITUDINAL (X) - FULL LENGTH BANDS	11550	40
S04B	T25	150	TOP, LONGITUDINAL - WEST OF THE VOID	8188	42
S04C	T25	150	TOP, LONGITUDINAL - EAST OF THE VOID	3925	21
S05	T12	250 x 250	4-LEGGED LINKS @ 250, END 1500 EACH SIDE	2040	1848
S06	T12	300 x 300	2-LEGGED LINKS @ 300, MID ZONE	2040	384
S07	T12	250 x 250	4-LEGGED LINKS @ 250 THROUGHOUT THE PAD	2040	154
S08	T25	-	TRIMMERS, 5-T25 EACH SIDE/FACE/DIRECTION	3400	80
S09	T16	150	HOOPS, 3 LAYERS EACH FACE	5984	12
S10	T16	150	RADIAL BARS	1880	136
S11	T12	150	CLOSED LINKS @ 150 IN THE 1200 ANNULUS	2640	68
S12	T25	-	6-T25 TOP + 6-T25 BOTTOM IN THE 1200 THICKENING	4800	12
S13	T12	150	CLOSED LINKS @ 150	3340	18
S14	T25	-	6-T25 EACH FACE, TOP AND BOTTOM, 900 BAND	6050	24
S15	T25	-	4-T25 EACH FACE AT 45 DEG	4000	16
S16	T25	-	4-T25 TOP AND BOTTOM, 1200 BAND	7500	8

BAR MARK SCHEDULE - MAT FOUNDATION 600 AND SUMP PIT

MARK	BAR	SPACING	LOCATION	CUT mm	No.
F01	T16	150	BOTTOM, LONGITUDINAL (X)	11350	82
F02	T16	150	BOTTOM, TRANSVERSE (Y)	6100	147
F03	T16	150	TOP, LONGITUDINAL (X)	11350	82
F04	T16	150	TOP, TRANSVERSE (Y)	6100	147
F05	T12	250 x 250	LINK GRID 250 x 250 THROUGHOUT	1710	2112
F06	T16	150	EDGE U-BARS, ALL FREE EDGES	1780	376
F07	T16	150	WALL STARTERS W1-W4, EACH FACE	2335	720
F08	T20	150	WALL STARTERS W6/W7, EACH FACE	2535	132
F09	T12	150	WALL STARTERS W5, EACH FACE	1835	66
F10	T20	-	SUMP-PIT OPENING TRIMMERS	3100	32
F11	T16	150	VERTICAL, EACH FACE	2300	96
F12	T16	150	HORIZONTAL RINGS, EACH FACE	8000	22
F13	T16	150	BOTH WAYS, EACH FACE	2000	56

DESIGN BASIS - THIS SHEET

GOVERNING COMBINATION COMB 103 BLAST, w = 448.15 kPa
383.00 blast (344.7 x 1.111 DLF) + 40.65 engineered cover + 2.00 SIDL + 22.50 self weight.
NO LIVE LOAD ON THE ROOF AT BLAST - IS 4991 Cl. 11.2. Static ULS 101 = 127.7 kPa, so blast governs 3.51 : 1.

ONE-WAY ACTION IS NOT A CHOICE, IT IS FORCED

Aspect 20.8 / 5.0 = 4.2 and IS 456 Table 26 is tabulated only to ly/lx = 2.0, so two-way action cannot be claimed. Mp depends on the 5000 clear width alone - which is why lengthening the box (M1) was structurally free, and why widening 5.0 -> 6.0 m would raise the roof moment 44 %.

ROOF (PRESSURE) SLAB 900 d = 900 - 75 - 12.5 = 812.5
Mp = w Ln2 / 16 = 448.15 x 25 / 16 = 700.2 kNm/m ; Mu,lim = 3841 kNm/m ; Ast,req = 1632 mm2/m
PROVIDED T25 @ 150 EF EW = 3272 mm2/m -> Mu 1362.4 kNm/m, UTILISATION 51 %, xu/d = 0.139 << 0.46
SHEAR V at d = 756.3 kN/m ; tv 0.931 ; tc 0.450 ; tc,max 3.70 -> T12 4L @ 250 end 1500, T12 2L @ 300 mid
NATURAL PERIOD fl = 74.9 Hz, T = 13.4 ms, td/T = 10 to 100 -> QUASI-STATIC. This is what licenses a static analysis at all.

CANTILEVER PAD 1840 (FINDING F2) M(root) = 448.15 x 1.8402 / 2 = 758.6 kNm/m - MORE THAN THE MIDSPAN Mp
Ast,req TOP 1772 mm2/m < 3272 provided -> 56 %. V(root) 824.6 kN/m -> T12 4L @ 250 throughout the pad.

MAT FOUNDATION 600 d = 600 - 75 - 8 = 517
CASE 1 net uplift 31.1 kPa over 5000 -> Mp 48.6 kNm/m, nominal.
CASE 2 *** GOVERNS *** a 3.0 m red-bole band removed at the worst position, q = 404.9 kPa blast bearing
M = q L2 / 12 = 303.7 kNm/m ; Ast,req 1115 mm2/m
PROVIDED T16 @ 150 EF EW = 1340 mm2/m -> Mu 362.8 kNm/m, UTILISATION 84 %
ONE-WAY SHEAR V at d = 398.0 kN/m ; tv 0.770 ; tc 0.375 -> T12 closed links on a 250 x 250 grid
BEARING 404.9 kPa on a presumptive 3240 kPa (IS 1904 hard rock) = 12.5 %. BEARING GOVERNS NOTHING.
PUNCHING IS 456 Cl. 31.6 NOT APPLICABLE - nothing bears on the mat; the headhouse walls bear on the ROOF.

OPENINGS 2 No. 1400 dia escape shafts, collar 250 RC, slab thickened 900 -> 1200 over a 600 annulus.
Circular is the right shape: hoop action and NO re-entrant stress concentration. This detail sets the 750 mm clearance rule, which is why a bay must be at least 2900 wide to hold an escape shaft.
1 No. 2800 x 3160 rectangular stair void - both re-entrant corners trimmed with S15 at 45 degrees.

OPEN ITEMS AFFECTING THIS SHEET A2 design GWT (-)2.000 is [ASSUMED] and carries two-thirds of the lateral load and all of the uplift - monsoon monitoring required. A4 subgrade modulus ks: the master requires BOTH 100 000 and 500 000 kN/m3 and only 100 000 exists in any model. M1 no STAAD result exists for any underground model - every design action above is a hand calculation. P3 blast capacity is not demonstrated; support rotation needs a non-linear SDOF check (IS 4991 Fig. 6 / Biggs) - Phase 3.

UNDERGROUND CBRN-HARDENED
BLAST-RESISTANT PROTECTIVE
STRUCTURE + SENTRY POST
PUNE, MAHARASHTRA
STRUCTURAL - PHASE 2 REV A + M1

NOTES

1. ALL DIMENSIONS ARE IN MM. ALL LEVELS ARE IN METRES RELATIVE TO FINISHED SITE GRADE 0.000, NEGATIVE DOWNWARDS.
2. WRITTEN DIM TO BE FOLLOWED ONLY. DRG NOT TO BE SCALED.
3. THIS DRG IS TO BE READ IN CONJUNCTION WITH ALL RELEVANT ARCH AND STR DRGS.
4. CONCRETE M35, w/c NOT MORE THAN 0.45. REINFORCEMENT Fe500D TO IS 1786:2008. Ld 40 PHI, LAPS 50 PHI, ALL STAGGERED SO THAT NOT MORE THAN 50 % ARE SPLICED AT ONE SECTION.
5. MAX BAR SPACING 150 IN BOTH CURTAINS - THIS IS AN EMP REQUIREMENT AND IS STRICTER THAN IS 456 Cl. 26.3.3.
6. DESIGN IS TO IS 456:2000 WITH IS 4991:1968 BLAST ACTIONS AND IS 3370 (Pt 2) CRACK CONTROL; SEISMIC TO IS 1893 (Pt 1):2016. IS 13920:2016 IS CHECKED: Cl. 10.4 BOUNDARY ELEMENTS ARE NOT TRIGGERED - SEE THE DESIGN BASIS PANEL.
7. CONTRACTOR TO CHECK AND VERIFY THE DRG BEFORE EXECUTION.

STRUCTURAL REINFORCEMENT DETAILING - ROOF SLAB & MAT FOUNDATION

CBRN HARDENED UG OPS ROOM

SHEET 06

Project number	01
Date	16 SEP 2026
Drawn by	SYN 01
Checked by	DR IR CHAUDHARI

STR006

Scale	1 : 100
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